UNIT - II

UNIFORM FLOW IN OPEN CHANNELS

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Open channel flow; comparison with pipe flow
- Geometric parameters; classification of channels and flow
- Velocity distribution of channel section
- Hydraulically efficient sections: rectangular, trapezoidal, triangular
- Energy and momentum correction factors

1. Introduction to Open Channel Flow

1.1 Definition

Open channel flow is the flow of a liquid with a **free surface exposed to atmosphere** (pressure on the surface = atmospheric). Flow is driven by **gravity** (component of weight along the slope).

- Examples: rivers, canals, drains, flumes, culverts, partially filled sewers and pipes.

1.2 Comparison between open channel flow and pipe flow

Basis	Open channel flow	Pipe flow
Free surface	Present, at atmospheric pressure	Absent; pipe flows full
Driving force	Gravity (bed slope)	Pressure difference
Hydraulic gradient line	Coincides with the water surface	Above the pipe axis, a pressure head
Cross-section	Varies: shape, depth and area change	Fixed (circular)
Roughness	Varies with depth, time (silt, vegetation)	Roughly constant
Flow area	A depends on depth y	Fixed
Analysis	More difficult (extra variable y)	Simpler

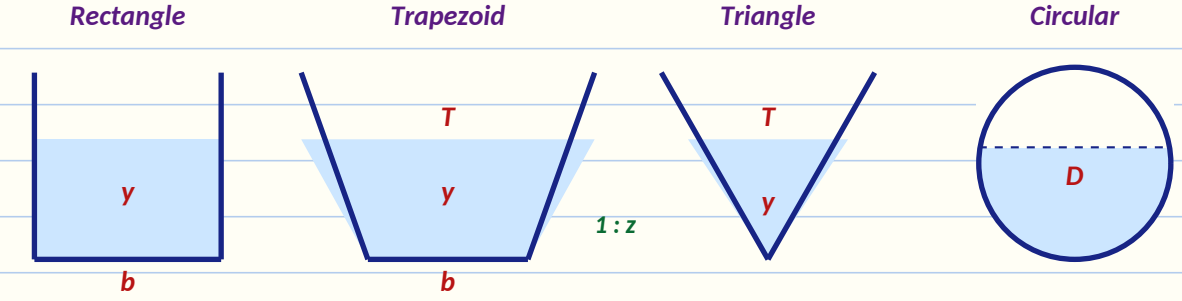
Design equation	Manning / Chezy	Darcy-Weisbach
Max velocity	Slightly below the surface	At the centre

2. Geometric Parameters of a Channel

2.1 Definitions

Term	Meaning
Depth of flow (y)	Vertical distance from the lowest point of the section to the free surface
Top width (T)	Width of the channel section at the free surface
Flow area (A)	Cross-sectional area of flow normal to the direction of flow
Wetted perimeter (P)	Length of the channel boundary in contact with water
Hydraulic radius (R)	$R = A / P$
Hydraulic depth (D)	$D = A / T$ (mean depth)
Section factor (Z)	$Z = A\sqrt{D} = A\sqrt{(A/T)}$; used for critical flow
Bed slope (S0)	Longitudinal slope of the channel bed = $\sin\theta \approx \tan\theta$
Side slope (z)	z horizontal to 1 vertical
Freeboard	Vertical distance from water surface to top of the bank

2.2 Formulas for common sections



Section	Area A	Wetted perimeter P	Top width T
Rectangle (b, y)	$b\,y$	$b + 2y$	b
Trapezoid (b, y, z)	$(b + z\,y)\,y$	$b + 2y\sqrt{(1 + z^2)}$	$b + 2\,z\,y$
Triangle (z, y)	$z\,y^2$	$2\,y\sqrt{(1 + z^2)}$	$2\,z\,y$
Circle (D, angle 2θ)	$(D^2/8)(2\theta - \sin 2\theta)$	$D\,\theta$	$D \sin\theta$

Example: Trapezoidal $b = 4$ m, $z = 1.5$, $y = 2$ m: $A = (4 + 3) \times 2 =$ **14 m²** ; $P = 4 + 2 \times 2 \times \sqrt{3.25} = 4 + 7.21 =$ **11.21 m** ; $R = 14 / 11.21 =$ **1.249 m** ; $T = 4 + 2 \times 1.5 \times 2 =$ **10 m** ; $D = 1.4$ m.

3. Classification of Open Channels

3.1 By origin

- **Natural channels** : rivers, streams, estuaries; irregular shape and roughness.
- **Artificial channels** : canals, flumes, drains, spillways, culverts; regular shape.

3.2 By cross-section and slope

- **Prismatic channel** : constant cross-section and bed slope along its length. **Non-prismatic**: varying section or slope.
- **Rigid boundary** (lined concrete) and **mobile boundary** (alluvial, erodible channel with sediment transport).
- **Shapes** : rectangular, trapezoidal, triangular, circular, parabolic, U-shaped, horse-shoe.

3.3 By bed slope

- **Mild slope** , **steep slope** , **critical slope** , **horizontal** , **adverse** (rising) slope.

4. Classification of Open Channel Flow



4.1 With respect to time

- **Steady flow** : depth, velocity do not change with time at a section ($\partial y/\partial t = 0$). **Unsteady flow** : they change (flood wave, surges).

4.2 With respect to space

- **Uniform flow** : depth, area, velocity same at every section along the channel. **Non-uniform (varied) flow** : they change along the length.
- Varied flow is of two types: **Gradually varied flow (GVF)** (depth changes slowly: backwater curve) and **Rapidly varied flow (RVF)** (depth changes abruptly over short length: hydraulic jump, flow over weir).

4.3 With respect to Reynolds number

Re = V R / ν (R = hydraulic radius)

- **Laminar**: Re < 500; **transition**: 500 to 2000 (some books: 500 to 12,500); **turbulent**: Re > 2000 (most channels are turbulent).

4.4 With respect to Froude number

Fr = V / √(g D) (D = hydraulic depth; for rectangle D = y)

- **Subcritical (tranquil)**: Fr < 1; **Critical**: Fr = 1; **Supercritical (rapid / shooting)**: Fr > 1.
- Fr is the ratio of **inertia to gravity force**. A small surface wave (celerity $c = \sqrt{g y}$) can travel **upstream only if flow is subcritical**.

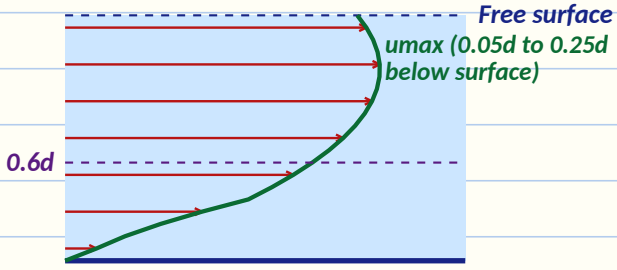
4.5 Summary

Type	Condition	Notes
Uniform steady	Sf = S0 = water surface slope	Normal depth yn
Gradually varied	y changes slowly	Backwater and drawdown curves
Rapidly varied	y changes abruptly	Hydraulic jump, drop, sluice gate

5. Velocity Distribution in a Channel Section

5.1 Pattern

- Because of friction at bed and banks and the free surface, velocity varies in both vertical and transverse directions. **Maximum velocity occurs below the free surface** (at 0.05 d to 0.25 d from the surface; deeper in narrow channels).



Mean velocity V:

V = u at 0.6 depth, or

V = (u0.2 + u0.8) / 2

V = 0.85 x surface velocity

5.2 Vertical velocity profile

- Approximately **logarithmic** (for wide channel): $u = (v/\kappa) \ln(y/y0)$, **with** $\kappa = 0.4$; **also power law** $u = u_{max} (y/d)^{(1/7)}$.

5.3 Mean velocity from measurement

Method	Mean velocity
Single point	V = u at 0.6 depth from the surface
Two-point	V = (u at 0.2d + u at 0.8d) / 2
Three-point	V = (u0.2 + 2 u0.6 + u0.8) / 4
Surface float	V = k x surface velocity, k ≈ 0.85 (0.8 to 0.95)

- **Instruments** : current meter (cup or propeller type), Pitot tube, floats, ADCP (acoustic Doppler).

5.4 Wide rectangular channel (laminar)

- $u = (g\ S0 / 2\nu)(2\ d\ y - y^2)$; the mean velocity = **$V = g\ S0\ d^2 / 3\nu$** and the maximum velocity at the surface = 1.5 V.

6. Uniform Flow Formulae

6.1 Conditions for uniform flow

- Depth, area, velocity and discharge are constant; **bed slope = water surface slope = energy slope (S0 = Sw = Sf)** .
- Force balance: **gravity component along the slope = shear resistance at wall** : $\rho\ g\ A\ S0\ L = \tau 0\ P\ L$.

$$\tau 0 = \rho\ g\ R\ S0$$

6.2 Chezy's formula (1769)

$$V = C\sqrt{(R\ S0)} \quad ; \quad Q = A\ C\sqrt{(R\ S0)}$$

- C = Chezy coefficient ($m^{1/2}/s$). Derived by equating $\tau 0 = k\ \rho\ V^2$ with $\tau 0 = \rho\ g\ R\ S0$.
- **Bazin** : C = 157.6 / (1 + m/√R); **Kutter** : empirical formula with n; **Manning**: C = R^(1/6) / n; **Powell** ; **Pavlovsky** .

6.3 Manning's formula (most used)

$$V = (1/n)\ R^{(2/3)}\ S0^{(1/2)} \quad ; \quad Q = (1/n)\ A\ R^{(2/3)}\ S0^{(1/2)}$$

- n = Manning's roughness coefficient ($s/m^{(1/3)}$).

Surface	n
Smooth cement / planed wood	0.010
Concrete, finished, brick	0.013 to 0.015
Unlined earth channel, clean	0.022 to 0.025
Natural stream with weeds	0.035 to 0.050

Rough rock / heavy vegetation	0.05 to 0.100

- **Conveyance K = (1/n) A R^(2/3)** so Q = K √S0. **Normal depth yn** = depth at which uniform flow occurs; solve A R^(2/3) = Q n / √S0 (section factor).

6.4 Solved Problems

Problem 1: Rectangular channel b = 3 m, y = 1.5 m, n = 0.015, S0 = 1/1000. Find V and Q.

- A = 4.5 m²; P = 3 + 3 = 6 m; R = 0.75 m; R^(2/3) = 0.8255.

V = (1/0.015) x 0.8255 x √0.001 = 1.74 m/s; Q = 4.5 x 1.74 = *7.83 m³/s*

Problem 2: Trapezoidal channel b = 4 m, z = 1.5, y = 2 m, n = 0.02, S0 = 0.0004. Find Q.

- A = 14 m²; P = 11.21 m; R = 1.249 m; R^(2/3) = 1.160.

V = (1/0.02) x 1.160 x 0.02 = 1.16 m/s; Q = 14 x 1.16 = *16.24 m³/s*

Problem 3: Find Chezy's C for R = 0.75 m, n = 0.015 and verify V (Problem 1).

- C = R^(1/6) / n = 0.9532 / 0.015 = 63.6; V = 63.6 x √(0.75 x 0.001) = 63.6 x 0.02739 = **1.74 m/s**. matches Manning.

Problem 4: Froude number for V = 2 m/s, y = 1.5 m (rectangular).

- Fr = 2 / √(9.81 x 1.5) = 2 / 3.836 = **0.52** (subcritical, tranquil).

Problem 5: Current-meter readings at 0.2d and 0.8d are 1.2 and 0.8 m/s; area is 5 m².

V = (1.2 + 0.8)/2 = 1.0 m/s; Q = 5 x 1.0 = 5 m³/s

7. Hydraulically Efficient Channel Sections

7.1 Concept

For a given area A, slope S0 and roughness n, the section that gives **maximum discharge** is the one with **minimum wetted perimeter P** (maximum hydraulic radius R). This is the **most economical / best hydraulic section**.

- Since Q = (1/n) A R^(2/3) S0^(1/2) = (1/n) A^(5/3) P^(-2/3) S0^(1/2). For constant A, Q is max when P is min.
- Among all shapes the **semi-circle** is the best, but it is difficult to construct; so rectangle,

trapezoid and triangle are used.

7.2 Rectangular section

- $A = b\,y$; $P = b + 2y = A/y + 2y$. Set $dP/dy = 0$: $-A/y^2 + 2 = 0$, so $A = 2y^2$.

Best rectangular section: $b = 2y$ and $R = y / 2$

- Width is twice the depth; depth of flow equals twice the hydraulic radius.

7.3 Trapezoidal section

- $A = (b + z\,y)\,y$; $P = b + 2y\sqrt{(1 + z^2)}$. Substituting $b = A/y - z\,y$, $P = A/y - z\,y + 2\,y\,\sqrt{(1 + z^2)}$. $dP/dy = 0$:

$$A = y^2\,[2\sqrt{(1 + z^2)} - z] \qquad b = 2\,y\,[\sqrt{(1 + z^2)} - z]$$

Best trapezoidal section: $R = y / 2$ (for every side slope z)

- For the best trapezoid the length of each sloping side equals half the top width (and equals b when $z = 1/\sqrt{3}$). With $z = 1/\sqrt{3}$ the section is half of a regular hexagon (side slope 60° to horizontal).
- The best side slope for a trapezoid: $z = 1/\sqrt{3} = 0.577$ (60°); then $b = 2y/\sqrt{3} = 1.155\,y$, $A = \sqrt{3}\,y^2 = 1.732\,y^2$, $P = 2\sqrt{3}\,y = 3.464\,y$.
- A semi-circle drawn with centre at the water-surface mid-point and radius y touches all three sides (bed and two banks).

7.4 Triangular section

- $A = z\,y^2$; for a given A, $P = 2\,y\,\sqrt{(1 + z^2)} = 2\,\sqrt{(A)}\,\sqrt{((1 + z^2)/z)}$. Minimum when $d/dz[(1 + z^2)/z] = 0$, so $z = 1$.

Best triangular section: $z = 1$ (side slopes 45°, vertex angle 90°); $R = y / (2\sqrt{2}) = 0.354\,y$; $A = y^2$

7.5 Circular section

- Max discharge when flow depth $y = 0.95\,D$ (θ about 154°); max velocity when $y = 0.81\,D$. At $y = D$ (full), Q is slightly lower than max, hence an enclosed pipe is not "best" when full.

7.6 Comparison for the same area $A = 8\ m^2$ (best sections)

Section	Depth y	Wetted perimeter P	R
Rectangle ($b = 2y$)	2.00 m	8.00 m	1.00 m
Trapezoid ($z = 0.577$)	2.15 m	7.44 m	1.075 m
Triangle ($z = 1$)	2.83 m	8.00 m	1.00 m
Semi-circle (reference)	$r = 2.26$ m	7.09 m	1.13 m

- The half-hexagon trapezoid has the least perimeter among practical sections.

7.7 Solved Problems

Problem 6: Design the best rectangular section to carry Q = 10 m^3/s, n = 0.015, S0 = 1/2500.

- b = 2y; A = $2y^2$; R = y/2. Q = (1/n) A R^(2/3) √S0 = 66.67 x $2y^2$ x 0.63 y^(2/3) x 0.02 = 1.68 y^(8/3).

1.68 y^(8/3) = 10 y^(8/3) = 5.95 *y = 1.95 m*, b = 3.90 m; V = 10 / 7.62 = 1.31 m/s

Problem 7: Best trapezoidal section for A = 8 m^2 (z = 1/√3).

- A = 1.732 y^2 = 8, so y = 2.149 m; b = 1.155 y = 2.48 m; P = 3.464 x 2.149 = 7.44 m; R = y/2 = 1.07 m.

Problem 8: Triangular section, best form, Q = 2 m^3/s, n = 0.02, S0 = 0.001. Find y.

- A = y^2; R = 0.354 y; Q = (1/0.02) x y^2 x (0.354 y)^(2/3) x 0.0316 = 50 x 0.0316 x 0.5 y^(8/3) = 0.79 y^(8/3).

0.79 y^(8/3) = 2 y^(8/3) = 2.53 y = 1.41 m (top width T = 2y = 2.82 m)

8. Energy and Momentum Correction Factors

8.1 Why needed

- Velocity is not uniform over the cross-section, so the **true kinetic energy flux and momentumflux** differ from those computed using the mean velocity V.

8.2 Kinetic energy correction factor (Coriolis coefficient α)

$$\alpha = (1 / A V^3) \int u^3 \, dA \quad (\alpha \geq 1)$$

- Kinetic energy per unit weight = $\alpha V^2 / 2g$. Specific energy: $E = y + \alpha V^2 / 2g$.

8.3 Momentum correction factor (Boussinesq coefficient β)

$$\beta = (1 / A V^2) \int u^2 \, dA \quad (\beta \geq 1)$$

- Momentum flux = $\beta \rho Q V$. Momentum equation: F = ρ Q (β2 V2 - β1 V1).
- Typical values: **α = 1.03 to 1.36**, **β = 1.01 to 1.12**; α > β always (α - 1 ≈ 3(β - 1)). Uniform velocity: α = β = 1.
- Laminar flow in a pipe: α = 2, β = 4/3; laminar between plates: α = 54/35, β = 6/5.

8.4 Power-law profile

u = u_max η^m (η = y/d): α = (1 + m)³ / (1 + 3m) ; β = (1 + m)² / (1 + 2m)

$m = 1/7$: $\alpha = (8/7)^3 / (10/7) = 1.045$; $\beta = (8/7)^2 / (9/7) = 1.016$.

8.5 Solved Problem 9

The section is divided into 4 equal areas where mean velocities are 1.0, 1.5, 2.0 and 1.5 m/s. Find α and β.

- Mean $V = (1.0 + 1.5 + 2.0 + 1.5)/4 = 1.5$ m/s.

$$\alpha = \Sigma u^3 / (n V^3) = (1 + 3.375 + 8 + 3.375) / (4 \times 3.375) = 15.75 / 13.5 = *1.167*$$

$$\beta = \Sigma u^2 / (n V^2) = (1 + 2.25 + 4 + 2.25) / (4 \times 2.25) = 9.5 / 9 = *1.056*$$

9. Design Aspects of Channels

9.1 Permissible velocity

- Velocity must be high enough to **prevent silting** (minimum velocity, about 0.6 to 0.9 m/s) and low enough to **prevent scouring** (maximum permissible velocity, depends on boundary material).

Boundary material	Approx. permissible velocity (m/s)
Fine sand	0.45
Sandy loam	0.55
Silt loam / ordinary firm loam	0.6 to 0.75
Stiff clay	1.1
Fine gravel	1.5
Hard pan, lined concrete	1.8 to 3 or more

9.2 Side slopes (z) and freeboard

- Typical side slopes (horizontal : vertical): **rock 0 to 0.25; stiff clay 0.5 to 1; firm earth 1 to 1.5; loose earth / sandy soil 2 to 3.** Freeboard is typically 0.3 to 1.0 m in canals.

9.3 Tractive force (shear) approach

$$\tau_0 = \gamma R S_0 \quad \text{(average boundary shear stress)}$$

- On the sloping sides the maximum shear is about 0.76 γ y S0; on the bed it is about γ y S0. Channel is stable if τ0 < permissible tractive force of the soil.

9.4 Composite roughness and compound sections

- **Composite roughness** (Horton / Einstein): different n_i over parts P_i of the perimeter:

$$n_e = [\ \Sigma P_i n_i^{1.5} / P\ ]^{(2/3)}$$

- **Compound section** (river with flood plains): divide into sub-areas, compute Q for each (using its own A, R and n) and add.

9.5 Circular channels partly full

- Flow depth ratio y/D: **Q is max at y/D = 0.95** and **V is max at y/D = 0.81** . At half-full, R = D/4 (same as full) so V is the same as full pipe; **Q(half) = 0.5 Q(full)**.

10. More Solved Problems

Problem 9: Normal depth by trial

Rectangular channel b = 4 m, Q = 8 m³/s, n = 0.015, S0 = 0.0004. Find yn.

- Required A R^(2/3) = Q n / √S0 = 8 x 0.015 / 0.02 = **6.0** .

y (m)	A	P	R	A R^(2/3)
1.5	6.0	7.0	0.857	5.41
1.6	6.4	7.2	0.889	5.92
1.62	6.48	7.24	0.895	6.02

Normal depth yn ≈ 1.617 m (V = 8 / 6.47 = 1.24 m/s)

Problem 10: Composite roughness

Rectangular channel b = 4 m, y = 2 m. Bed n = 0.015; side walls n = 0.025. Find equivalent n.

- P1 (bed) = 4 m; P2 (two sides) = 4 m; P = 8 m. n_e = [(4 x 0.015^1.5 + 4 x 0.025^1.5) / 8]^(2/3) = (0.002895)^(2/3).

n_e = *0.0203*

Problem 11: Circular culvert half full

D = 1.2 m, depth y = 0.6 m (half full), n = 0.013, S0 = 0.002. Find V and Q.

- A = π D² / 8 = 0.5655 m²; P = π D / 2 = 1.885 m; R = D/4 = 0.3 m; R^(2/3) = 0.448.

V = (1/0.013) x 0.448 x √0.002 = 1.54 m/s; Q = 0.5655 x 1.54 = *0.87 m³/s* (half of the full-pipe 1.74 m³/s)

Problem 12: Required bed slope

Trapezoidal canal b = 5 m, z = 1, y = 2 m, n = 0.025 must carry Q = 20 m³/s. Find S0.

- A = 14 m²; P = 5 + 2 x 2 x 1.414 = 10.66 m; R = 1.314 m; R^(2/3) = 1.20; V = 20/14 =

1.43 m/s.

$S0 = (V\,n / R^{(2/3)})^2 = (1.43 \times 0.025 / 1.20)^2 = 8.87 \times 10^{-4}$ (about 1 in 1130)

Problem 13: Discharge of best trapezoidal section

$y = 1.5$ m, $z = 1/\sqrt{3}$, $n = 0.02$, $S0 = 0.0005$ (best section). $A = 1.732\,y^2 = 3.90$ m²; $R = y/2 = 0.75$ m.

$Q = (1/0.02) \times 3.90 \times 0.8255 \times 0.02236 = *3.60\ m^3/s*$

Problem 14: Boundary shear stress

Using Problem 2: $R = 1.249$ m, $S0 = 0.0004$.

$\tau 0 = 1000 \times 9.81 \times 1.249 \times 0.0004 = 4.9\ N/m^2$ (compare with the soil's permissible tractive force)

Problem 15: Reynolds and Froude numbers

Rectangular channel, $b = 2$ m, $y = 0.5$ m, $V = 3$ m/s, $\nu = 10^{-6}$ m²/s.

- $R = A/P = 1.0/3.0 = 0.333$ m; $Re = V R / \nu = 3 \times 0.333 / 10^{-6} = 10^6$ (turbulent). $Fr = 3 / \sqrt{(9.81 \times 0.5)} = 1.35$ (> 1, supercritical).

11. Derivations and Discharge Estimation

11.1 Derivation of Chezy's formula

- Consider uniform flow in a length L. **Forces**: weight component $W \sin\theta = \rho g A L S0$ (driving); **resisting force** $= \tau 0\, P\, L$.
- For uniform flow they balance: $\rho g A L S0 = \tau 0 P L$, so $\tau 0 = \rho g R S0$.
- Experiments show resistance varies as the square of velocity: $\tau 0 = k \rho V^2$.

$k \rho V^2 = \rho g R S0 \quad V = \sqrt{(g/k)}\,\sqrt{(R S0)} = C\sqrt{(R S0)} \quad$ where $C = \sqrt{(g/k)}$

11.2 Velocity-area method of discharge measurement

- Divide the cross-section into strips of width w; measure depth d and mean velocity v in each strip with a current meter. $Q = \Sigma (w\,d\,v)$.

Example: 4 strips of width 2 m; depths 0.5, 1.0, 1.5, 1.0 m; mean velocities 0.4, 0.7, 0.9, 0.6 m/s.

$Q = 2 \times (0.5 \times 0.4 + 1.0 \times 0.7 + 1.5 \times 0.9 + 1.0 \times 0.6) = 2 \times 2.85 = 5.7\ m^3/s$

11.3 Slope-area method (floods)

- Use two sections of a reach: Manning with the average conveyance, $Q = K\sqrt{Sf}$, where K

= (1/n)(A R^(2/3)) average of sections, and Sf found from energy equation between the sections.

11.4 Chezy vs Manning

Basis	Chezy	Manning
Formula	V = C √(R S0)	V = (1/n) R^(2/3) S0^(1/2)
Coefficient	C varies with R and roughness	n depends mainly on roughness only
Use	Pipes and channels, theory	Standard in practice for channels
Relation	C = R^(1/6) / n	n = R^(1/6) / C

12. Quick Revision and Important Questions

Open channel: free surface, gravity-driven. R = A/P; D = A/T; Fr = V/√(gD). Manning: V = (1/n)R^(2/3)S^(1/2). Best sections: rectangle b = 2y; trapezoid half-hexagon, R = y/2; triangle z = 1. $\alpha = \int u^3 dA/(AV^3)$; $\beta = \int u^2 dA/(AV^2)$.

- Compare open channel flow with pipe flow. Classify open channel flow with examples. (5 M)
- Explain the velocity distribution in an open channel and the methods of finding mean velocity. (5 M)
- Derive Chezy's formula and state Manning's formula. (5 M)
- Derive the conditions for the most economical (i) rectangular (ii) trapezoidal (iii) triangular section. (10 M)
- Define energy and momentum correction factors; derive them. (5 M)
- Problems: discharge for given section, design of best section, Froude number classification.